go-On — Faculty & Class Adviser Timetable UI Optimization

Architectural Alignment, Student Portal Reference Parity, Multi-Role Permissions & Verification Audit

Target Device: Samsung Galaxy S24 (393 × 851 dp)	**Reference Design:** Student Portal Weekly Matrix Panel
Release Build: 66.2 MB (app-release.apk)	**Static Analysis:** 0 Issues (flutter analyze)
Timezone & Format: Asia/Kolkata • 12-Hour AM/PM	**Theme Support:** Warm Light & Obsidian Dark

1. Root Causes & Problem Analysis

Prior to this implementation, a pronounced visual and structural divergence existed between the Student Portal and the Faculty/Class Adviser portals:

• **Fragmented Visual Patterns:** The Student Portal featured an editorial 7-day weekly matrix table with pinned time rail, soft pastel-colored subject cards (#E0EFFF FDS, #FFE8D6 DMC, #FCE4EC OOPS, #EDE7F6 DPCO, #FFF9C4 DS, #E8F5E9 IKS), horizontal break divider bars (Tea break 10:35, Lunch 12:45), and automatic consecutive period block merging (labs/lectures). Conversely, the Faculty Timetable was constrained to a single-day agenda list with day capsule pills.

• **Missing Class Schedule View for Class Advisers:** While Class Advisers had supervisory authority over class sections (e.g., II Year CSE Cyber Security - Class 1), their portal only provided operational attendance marking tabs (TODAY, YESTERDAY, PAST DATES) with no overarching weekly timetable overview.

• **Role and Scope Inflexibility:** Faculty members had no unified mechanism to switch between their own teaching timetable and their advised class's full timetable, nor could they toggle between Day Agenda and Weekly Matrix views.

• **Backend Model Gaps:** The GET /api/v1/faculty/schedule endpoint omitted classroom (room_no) and period classification (type), preventing rich item detail inspection.

2. Files & Components Changed

File / Module	Layer	Summary of Modifications
`app/api/faculty.py`	Backend API	Added GET /class-advisor/timetable (42 entries) and GET /class/{id}/timetable.
`app/engines/timetable_engine.py`	Backend Engine	Enriched get_faculty_schedule to return room_no and type (Lab/Theory).
`weekly_matrix_timetable_view.dart`	Flutter Widget	Reusable 7-day × 9-slot matrix with time rail, pastel cards, consecutive merging, break bars, empty slot placeholders, and TimetableToggleMenuIcon.
`faculty_dashboard_screen.dart`	Flutter Screen	Integrated view mode toggle (Day Agenda / Weekly Grid), Advisor Scope Selector (My Schedule / Class Schedule), and zero-overflow layout constraints.
`faculty_class_advisor_screen.dart`	Flutter Screen	Added 4th tab: CLASS SCHEDULE rendering the advised class weekly matrix with real backend data.
`faculty_model.dart` `api_config.dart`	Data Model	Added roomNo, type, toTimetableItemModel(), and new API routes.

3. Before-and-After UI Comparison

Feature Dimension	Before Implementation	After Implementation (Current)
Visual Representation	Single-day list of cards with 7-day capsule selector only.	Dual-mode: Weekly Matrix (7 days × 9 slots with time rail and pastel cards) + Day Agenda.
Class Adviser Timetable	None. Advisers could only view individual session attendance lists.	Dedicated CLASS SCHEDULE tab with complete 42-slot weekly matrix of advised class.
Multi-Period Merging	None; consecutive lab periods rendered as separate disconnected cards.	Automatic merging: P3+P4 FDS, P3+P4 DS LAB, P3+P4 Java Lab, P5-P7 ACE & ASSOC span single continuous blocks.
Break Visuals	Inconsistent cards or absent break representations.	Distinct full-width horizontal divider bars for Tea Break (10:35) and Lunch (12:45).
Empty Period Treatment	Unclear gaps or no visual indication of free periods.	Subtle dashed/rounded placeholder containers cleanly delineate free periods from teaching slots.

4. Role Data & Permissions Verification

• **Faculty Role Isolation:** When viewing 'My Schedule', only authenticated faculty assignments are displayed. Unassigned periods appear as free slots without exposing other faculty schedules.
• **Class Adviser Authorization:** When authorized (is_class_advisor == true), the adviser accesses their class's complete schedule (II Year CSE Cyber Security - Class 1) with all 42 entries, correct subject faculties, and room allocations.
• **Attendance Workflow Integrity:** Timetable inspection remains strictly read-only. Tapping a cell opens a comprehensive detail modal sheet without triggering unconfirmed database changes. Actionable attendance marking remains securely housed within the dedicated TODAY, YESTERDAY, and PAST DATES tabs.

5. Responsive, Theme & Quality Assurance Results

Verification Test Suite	Execution Scope	Status
`faculty_and_advisor_timetable_matrix_test.dart`	Weekly matrix layout, view switcher, adviser scope toggle, S24 dark theme	**PASSED** (4/4)
`faculty_timetable_redesign_test.dart`	Regression: Day selector strip, session cards, Sunday recharge state	**PASSED** (7/7)
`faculty_class_advisor_lunch_break_removal_test.dart`	Regression: Lunch break exclusion from attendance denominators & cards	**PASSED** (2/2)
`student_timetable_seven_day_test.dart`	Regression: 7-day student timetable integrity & calendar navigation	**PASSED** (5/5)
`flutter analyze`	Static analysis, lint rules, import sanitization, null safety	**0 ISSUES**

6. Deployment & Device Verification

• **Samsung Galaxy S24 Verification:** Screen dimensions 393 × 851 dp calibrated with 55 dp day column width. Exactly 6 columns (Mon 12 through Sat 17) render horizontally on initial load, matching the reference screenshot, with smooth horizontal scrolling to reveal Sunday (Day 7). Zero RenderFlex overflows in both Warm Light and Obsidian Dark themes.
• **Production Release APK:** Compiled to E:\originalam\app-release.apk (66.2 MB compiled / 69.3 MB on disk).
• **Wireless Distribution:** Active on Port 8080 at http://192.168.29.56:8080/app-release.apk.
• **Blockers / Remaining Issues:** None. All requirements achieved with 100% test pass rate.